

ROZHEN OBSERVATORY, Bulgaria

WMO: 157260

Lat: 41.6950N

Lon: 24.7390E

Elev: 1750

StdP: 81.99

Time Zone: 2.00 (EUE)

Period: 01-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.9	-13.7	-20.9	0.7	-6.4	-18.7	0.9	-7.6	8.4	-7.2	7.9	-5.5	3.8	320	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.6	23.7	13.7	22.2	13.5	20.7	13.0	15.5	20.5	14.6	19.6	13.9	19.0	2.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.7	12.2	17.2	12.9	11.5	16.2	12.0	10.8	15.5	50.0	20.5	47.1	19.6	44.8	18.9	18.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.8	6.5	5.4	DB	-17.9	27.0	3.0	2.2	-20.1	28.5	-21.8	29.8	-23.5	31.0	-25.7	32.6	(4)
(5)			WB	-18.2	17.5	3.0	0.9	-20.3	18.2	-22.1	18.7	-23.7	19.3	-25.9	19.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.7	-3.5	-3.5	-0.2	3.8	9.1	12.7	15.0	15.4	10.9	7.1	3.0	-1.8	(6)
(7)		DBStd	8.06	5.32	5.47	4.83	4.03	3.88	3.70	3.23	3.13	3.87	4.73	5.43	5.21	(7)
(8)		HDD10.0	2118	418	377	316	191	64	15	3	2	34	113	217	367	(8)
(9)		HDD18.3	4628	677	611	574	437	286	172	113	100	223	349	461	626	(9)
(10)		CDD10.0	552	0	0	1	4	37	96	157	169	61	22	7	0	(10)
(11)		CDD18.3	21	0	0	0	0	0	3	8	9	1	0	0	0	(11)
(12)		CDH23.3	76	0	0	0	0	0	9	36	30	1	0	0	0	(12)
(13)		CDH26.7	7	0	0	0	0	0	1	5	1	0	0	0	0	(13)
(14)	Wind	WSAvg	2.7	2.8	2.8	3.0	2.7	2.7	2.6	2.6	2.5	2.4	2.5	2.8	2.7	(14)
(15)	Precipitation	PrecAvg	714	69	62	63	57	68	60	48	36	50	56	61	84	(15)
(16)		PrecMax	1066	149	150	131	129	144	137	130	97	143	143	126	170	(16)
(17)		PrecMin	433	9	16	18	17	37	22	5	8	10	7	8	7	(17)
(18)		PrecStd	130	32	35	30	24	24	26	27	22	30	32	27	39	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.8	10.3	13.3	17.6	20.4	24.2	26.9	26.4	22.6	19.9	17.3	11.1	(19)
(20)			MCWB	2.2	2.9	5.3	9.7	11.1	14.8	15.2	14.2	11.8	10.4	7.9	5.0	(20)
(21)		2%	DB	6.8	6.8	10.6	13.7	18.3	21.8	23.9	23.7	20.6	16.9	14.3	8.2	(21)
(22)			MCWB	0.8	1.2	4.0	6.4	10.5	13.7	13.5	13.8	11.7	9.6	6.8	3.1	(22)
(23)		5%	DB	4.6	5.0	8.1	11.3	16.6	20.1	22.0	22.1	18.3	14.9	11.6	6.2	(23)
(24)			MCWB	0.3	0.6	2.9	5.7	9.9	13.0	13.5	13.7	11.0	8.6	4.9	3.0	(24)
(25)		10%	DB	2.8	3.2	6.1	9.4	14.8	18.4	20.3	20.5	16.4	13.0	9.5	4.7	(25)
(26)			MCWB	0.2	0.2	2.0	4.6	9.1	12.4	13.1	13.0	10.4	7.8	4.2	1.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	3.7	6.2	10.2	12.7	16.2	17.0	16.2	13.6	12.0	9.9	6.5	(27)
(28)			MCDWB	6.0	7.4	11.7	17.3	17.6	21.1	23.6	21.1	19.1	18.2	15.1	8.2	(28)
(29)		2%	WB	2.5	2.3	4.6	7.5	11.4	14.9	15.6	15.3	12.5	10.6	8.0	5.0	(29)
(30)			MCDWB	4.6	4.8	9.0	12.4	16.1	19.7	20.7	20.3	17.8	14.7	11.3	6.8	(30)
(31)		5%	WB	1.3	1.5	3.5	6.2	10.5	13.9	14.6	14.5	11.7	9.7	6.7	3.6	(31)
(32)			MCDWB	3.4	3.9	7.2	10.3	15.0	18.6	19.3	19.9	16.4	13.3	9.6	5.4	(32)
(33)		10%	WB	0.3	0.6	2.4	5.2	9.7	13.0	13.8	13.8	11.0	8.8	5.5	2.3	(33)
(34)			MCDWB	2.1	2.7	5.5	8.4	13.8	16.9	18.5	19.3	15.0	11.9	8.0	4.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.7	6.1	6.8	6.7	6.8	6.9	7.6	7.6	7.0	6.6	6.3	5.4	(35)
(36)			MCDBR	7.9	8.4	8.7	9.1	8.8	8.6	9.4	9.1	9.2	8.8	8.3	7.1	(36)
(37)			MCWBR	4.9	5.2	4.9	5.1	5.0	4.3	4.3	4.0	4.7	5.1	5.3	4.6	(37)
(38)		5% WB	MCDWR	6.3	6.7	7.8	8.2	8.0	7.9	8.1	8.3	8.1	7.5	6.8	6.4	(38)
(39)			MCWBR	4.8	4.6	4.7	4.9	5.0	4.7	4.6	4.4	4.4	4.8	4.9	4.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.277	0.304	0.345	0.393	0.384	0.385	0.382	0.388	0.368	0.331	0.302	0.282	(40)	
(41)		taud	2.458	2.392	2.303	2.160	2.251	2.272	2.281	2.260	2.302	2.411	2.468	2.458	(41)	
(42)		Ebn at Noon	885	906	900	875	891	888	886	870	863	860	843	844	(42)	
(43)		Edh at Noon	75	94	116	144	135	133	130	129	115	91	74	69	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.74	2.56	3.59	4.60	5.61	6.13	6.50	5.95	4.43	3.04	2.02	1.54	(44)	
(45)		RadStd	0.23	0.31	0.33	0.51	0.40	0.53	0.47	0.39	0.34	0.31	0.19	0.23	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)		+0.45	N/A	N/A	N/A	+0.52	+0.50	-82	-147	+74	N/A

Nomenclature: See separate page